

# Halving hall

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming                             |
| Lesson          | Halving hall (lesson 7 of the unit)                                                                               |
| Lesson ID       | a1-l07                                                                                                            |
| Experience ID   | a1-l07                                                                                                            |
| Plan ID         | PLAN-AL-L07                                                                                                       |
| Content version | 756622cfa3c9                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=a1-l07">https://revise360.co.uk/experience.html?id=a1-l07</a> |

## Big question, outcomes and success criteria

**Big question** — How does a binary search work, and why is it so much faster?

### Prerequisites

The lesson before this one is a1-l06. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: a1-l01, a1-l02, a1-l03, a1-l04, a1-l05.

### Assessable outcomes

1. **a1-l07-o1** — understand the binary search algorithm
2. **a1-l07-o2** — know the condition a list must meet for a binary search to work
3. **a1-l07-o3** — understand which searching algorithm is quicker

## Vocabulary, resources and preparation

### Key vocabulary — Binary search, Midpoint

### Resources

|                   |                                              |
|-------------------|----------------------------------------------|
| SLIDES            | presentations/AL_L07_HalvingHall_Lesson.pptx |
| WORKSHEET         | worksheets/AL_L07_HalvingHall_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=a1-l07                    |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build APP\_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on AL_L07_HalvingHall_Worksheet.docx is filled in.                     |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find How it works.                                    | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. How it works

- Check the middle item of the list.
- If it matches, stop. If the target is larger, discard the lower half; if smaller, discard the upper half.
- Repeat on what is left until you find it or nothing remains.

**Challenge on this wall:** Describe a binary search in four steps of your own.

### 2. The one condition

- The list must be in order, or the algorithm cannot know which half to discard.
- If the data is unsorted, it must be sorted first, which takes time.
- Sorting once and searching many times is the situation where it pays.

**Challenge on this wall:** Explain why a binary search cannot be used on an unsorted list.

### 3. Why it is so fast

- Each comparison halves what is left.
- A list of 1,000 items takes about 10 checks; a million takes about 20.
- Doubling the list adds one comparison, not a thousand.

**Challenge on this wall:** Work out roughly how many checks a binary search needs for 64 items.

### 4. Comparing the two searches

- Linear: any order, simple, slow on large data.
- Binary: must be sorted, more complex, very fast on large data.
- For a handful of items the difference doesn't matter; for millions it decides whether the system works.

**Challenge on this wall:** Make a table comparing linear and binary search on three points.

### 5. Writing it down

- Keep two pointers: left at 0 and right at the last index.
- The midpoint is (left + right) divided by two, rounded down.
- Move left or right past the midpoint depending on the comparison.

**Challenge on this wall:** Write pseudocode for a binary search.

### 6. Answering binary search questions

- Always state that the list must be sorted: it is usually worth a mark.
- Describe the midpoint comparison and the discarding of half.
- If asked to show the steps, write down the midpoint checked at each stage.

**Challenge on this wall:** Plan a five-mark answer describing a binary search on a list of 15 names.

## Retrieval and the check for understanding

**Starter** — Think of a number between 1 and 100. How many yes/no questions would a friend need to guess it? What is the best question to ask first?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (a1 - 107 - s3 - c2), so the wording a pupil sees matches the wording you say.

**A list doubles in size. What happens to a binary search?**

**Correct:** It needs one extra comparison

### What a pupil might choose instead

- It takes twice as long
- It takes four times as long
- Nothing changes

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** That is why it copes with enormous data.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

|           |   |                                                                                                                           |
|-----------|---|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | 8 | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | 6 | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                           | ACTIVITIES | RECORD ON PAPER                                 |
|-------|-----------------------------------|------------|-------------------------------------------------|
| 1     | How it works                      | 2          | A binary search works by...                     |
| 2     | The one condition                 | 2          | A binary search requires...                     |
| 3     | Why it is so fast                 | 2          | A binary search is fast because...              |
| 4     | Comparing the two searches        | 2          | Linear and binary searches differ because...    |
| 5     | Writing it down                   | 2          | Pseudocode for a binary search...               |
| 6     | Answering binary search questions | 2          | To answer a binary search question, I should... |
| final | Final challenge: halve it         | 2          | the answer to the station's challenge           |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                                                                                               | MARKS | WHERE                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------|
| a1-107-x1 | Describe how a binary search works.                                                                                                    | 4     | worksheet, Exam practice a); deck, Exam practice slide |
| a1-107-x2 | State the condition a list must meet before a binary search can be used.                                                               | 1     | worksheet, Exam practice b); deck, Exam practice slide |
| a1-107-x3 | A sorted list contains 1,000 items. Compare the number of comparisons needed by a linear search and a binary search in the worst case. | 4     | worksheet, Exam practice c); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/a1-107.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Which item does a binary search look at first?** — The first item; The last item
- **Why must the list be sorted?** — Because it is faster to read; Because sorted lists use less memory
- **When is sorting first, then binary searching, worth the effort?** — When searching once; When the list is tiny
- **A sorted list has 1,000,000 items. Roughly how many checks does a binary search need?** — 1,000,000; 500,000
- **A list doubles in size. What happens to a binary search?** — It takes twice as long; It takes four times as long

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- How a binary search works
- Why the list must be sorted
- Comparing linear and binary searches

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is a1-108, Bubble bay. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

**If the internet or the devices are unavailable**

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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